An FSO-Based Drone Charging System for Emergency Communications

Di Wu, *Student Member, IEEE*, Xiang Sun, *Member, IEEE*, and Nirwan Ansari, *Fellow, IEEE*

***Abstract*—In a disaster struck area (DSA), macro base stations (MBSs) are usually damaged, and thus the wireless network becomes dysfunctional. To efficiently recover the communications, drone mounted base stations (DBSs) are deployed to relay data between the mobile users (MUs) in a DSA and working MBSs in the proximity of the DSA. However, a DBS may be deployed far away from a working MBS, thus limiting the backhaul link capacity between the DBS and the MBS. Also, the hovering time of current drones is limited, and thus caps the usage of DBSs. In order to increase the backhaul link capacity and prolong the hovering time of a DBS, we propose to apply free space optics to enable an MBS to simultaneously transmit data streams and energy to a DBS with high efficiency. The problem of jointly optimizing the DBS placement as well as the access link bandwidth allocation is formulated to maximize the hovering time of a DBS and guarantee the data rate requirements of the MUs. The joint bandwidth allocation DBS placement (TWIST) algorithm is proposed to solve the problem. The performance of TWIST is demonstrated through simulations.**

***Index Terms*—Drone, drone placement, emergency communications, free space optics, wireless charging.**

I. Introduction

OWING to the overexploitation of earth's resources, natural disasters (such as earthquake and tsunami) are frequently struck in recent years according to the research from Globe Change Data Lab [1]. Wired and wireless communications infrastructure (such as wireless access points) and related power systems could be severely damaged owing to natural disasters, thus leading to the whole communications network completely unavailable in a disaster struck area (DSA). Accordingly, the mobile users (MUs) in a DSA are unable to reach out to the people that are out of the DSA that may hinder, for example, the rescue teams to search for the victims and schedule evacuation paths [2]. Thus, how to quickly establish an emergency communications in a DSA is critical and has drawn much attention recently [3].

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Di Wu and Nirwan Ansari are with the Advanced Networking Laboratory, Department of Electrical and Computer Engineering, New Jersey Institute of Technology, Newark, NJ 07102 USA (e-mail: dw245@njit.edu; nirwan.ansari@njit.edu).

Xiang Sun is with the Smart Edge Computing and Networking Laboratory, Department of Electrical and Computer Engineering, University of New Mexico, Albuquerque, NM 87131 USA (e-mail: sunxiang@unm.edu).

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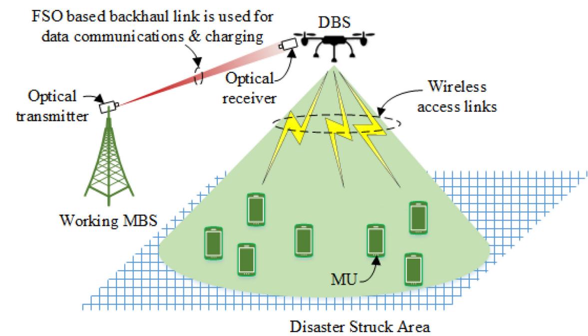

Fig. 1. The SoarNet architecture.

Deploying drone mounted base stations (DBSs) over DSAs could establish emergency wireless networks rapidly and efficiently. Specifically, a DBS can quickly fly over a DSA after a natural disaster and relay data between the MUs in the DSA and a macro base station (MBS), which is outside the DSA and works properly [4]. Network providers can flexibly deploy and adjust the DBS placement based on various conditions. Also, a DBS can likely provide line of sight (LoS) wireless links to the MUs as well as the working MBS, and thus the pathloss may potentially be reduced and the network capacity may be increased accordingly [5], [6].

Although a DBS can be quickly and efficiently placed over a DSA to set up emergency wireless communications, there are still some challenges. First, the DBS has to be placed over a DSA such that the distance between the DBS and the MUs in the DSA is close enough to provision high access link data rates. However, in a large DSA, the DBS has to be deployed far away from the working MBS, and thus the capacity of the backhaul link between the DBS and the working MBS may be limited. Accordingly, a bottleneck may be generated on the backhaul link when the MUs download data from the working MBS via the DBS. Second, current commercial drones, which are normally powered by portable batteries, have the maximum flying time around 30 minutes. That is, a DBS has to return to the control station for every 30 minutes of flight time and to be charged for several hours. The limited flying time hinders the usage of applying drones in emergence communications. In order to significantly improve the backhaul link capacity while expanding the battery life of a drone, we propose to leverage free Space optics as backhaul and energizer for drone assisted Networking (SoarNet) [7], [8] to provision emergency communications. As shown in Fig. 1, an optical transmitter is embedded into an MBS and emits an optical beam to a DBS. The emitted optical beam is used to transmit both data and energy to the DBS. An optical receiver is mounted on the DBS to receive the optical beam. The optical receiver is made up of two parts, i.e., a solar panel and an FSO receiver. The solar panel is used to obtain the optical energy to provide the DBS with extra energy in order to prolong its battery life. The FSO receiver is used to retrieve the data carried by the optical beam and transmit the received data to the MUs via the RF transmitter. Note that an FSO link has been demonstrated to provision a high link capacity over a long distance between two endpoints [9]. Meanwhile, it has been shown that optical beams, which are highly directional, can transmit/transfer energy more efficiently than traditional energy harvesting (such as RF and solar energy harvesting) [10], [11]. Therefore, SoarNet can simultaneously transmit the data streams to the DBS at high speed and charge the DBS with high efficiency.

Based on the proposed SoarNet architecture, a new DBS placement has to be designed. Different from several existing DBS placement algorithms in drone assisted mobile networks, which aim to maximize the network performance, we design a jointly optimized DBS placement and bandwidth allocation method to maximize the hovering time of the DBS and ensure the MUs' QoS in terms of the data rate requirements. Our main contributions are summarized below. 1) We propose to leverage the SoarNet architecture for emergency communications by applying FSO to simultaneously transmit the data streams and charge the DBS with high efficiency. 2) We formulate the problem of jointly optimizing the access link bandwidth allocation and 3-D DBS placement in SoarNet to maximize the hovering time of the DBS, and at the same time to satisfy the MUs' data rate requirements in the DSA. We design the joint bandwidth allocation DBS placement (TWIST) algorithm as a solution to the proposed problem. 3) We validate the performance of TWIST via extensive simulations.

The rest of this paper is organized as follows. Related works of drone assisted wireless networks are briefly summarized in Section II. The data rate models and FSO based energy charging model are illustrated in Section III. The problem of jointly optimizing the access link bandwidth allocation and 3-D DBS placement is illustrated in Section IV, and the proposed TWIST algorithm is provided in Section V. Simulation results are illustrated and analyzed in Section VI. We give a brief conclusion about our work in Section VII.

II. Related Works

Many researchers have explored the DBS placement and resource management methods in drone assisted mobile networks. Yaliniz *et al.* [12] investigated a 3-D DBS deployment algorithm that can maximize the number of served MUs, where an MU is able to be served by a DBS with favorable pathloss. In order to optimize the DBS deployment over a hotspot, Sun and Ansari [13] designed a novel DBS placement and user association method that can maximize the spectrum efficiency of the hotspot. By placing several DBSs above a DSA, Wu *et al.* [14] proposed a cooperative DBS aided wireless network architecture, where DBS-to-DBS communications is used to balance the traffic load of different backhaul links. They designed a wireless channel assignment and multiple DBS deployment method to maximize the network throughput.

Applying FSO as the backhaul solution to improve the achievable data rate of the backhaul link in drone mobile networks has recently been proposed [15]. Sun *et al.* [16] designed a 3-D DBS deployment algorithm that maximizes the spectrum efficiency of the network, where the backhaul link is implemented as an FSO link. However, they assumed that the FSO based backhaul link can provide sufficient link capacity to meet the data rate requirements of the access links between a DBS and the MUs. Fawaz *et al.* [17] applied a drone, which is equipped with an FSO transceiver, to relay traffic between, for example, two ground small cells, which do not satisfy the LoS criteria to maintain an FSO link. Alzenad *et al.* [18] explored an FSO-based vertical wireless communications architecture, where a number of drones (each of which are equipped with multiple FSO transceivers) interconnected together to establish a drone relay network to relay traffic between geographically distributed MBSs and a gateway via FSO links. Based on the proposed FSO-based wireless communications architecture, Gu *et al.* [19] proposed a method to agilely alter the routing paths such that the overall throughput is maximized and the total power consumption of the drones is minimized.

To extend the hovering time of drones, Alsharoa *et al.* [20] proposed that drones mounted with solar panels can collect solar energy, thus extending their hovering time. To avoid taking extra load for charging, Shin *et al.* [21] proposed to deploy movable wireless charging stations to charge drones. That is, movable wireless charging stations can be placed under drones to conduct wireless energy charging, and drones can continuously provide service to MUs while charging. Sang-Won *et al.* [22] designed a two receiving coils charging station to increase the wireless charging efficiency. In comparison to the traditional RF wireless charging technologies, using optical wireless charging can achieve much higher efficiency [23]. Our previous works have designed and illustrated the SoarNet architecture to simultaneously transmit energy and data from an MBS to a DBS [7]. However, how to jointly optimize the bandwidth assignment and the DBS deployment to maximize the hovering time of the DBS is still unveiled.

III. System Model

Assume that one DBS is placed over a DSA. Without loss of generality, the DSA is divided into a number of equal-sized locations. Let $\boldsymbol{I}$ and i be the set and index of these locations, respectively. In addition, let $\boldsymbol{J}$ be the set of MUs (with j to index each MU) that are located in the DSA. With the support of satellite technologies, we assume that the size of a DSA can be estimated. Also, the locations of different base stations are maintained by the network providers. By having such information, the distance between the locations in the DSA and the MBS can be estimated.

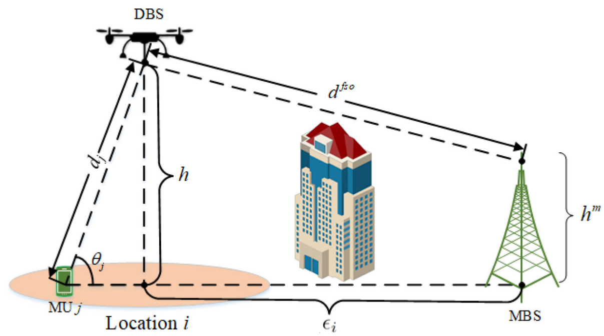

Fig. 2. Illustration of the pathloss model.

A. Access Link Communications Model

Denote z_i as a binary variable to imply the horizontal deployment of the DBS, i.e., if the DBS is placed over location i, $z_i = 1$; otherwise, $z_i = 0$. Thus, the horizontal distance between the DBS and MU j is

$$l_j = \sum_{i \in \boldsymbol{I}} z_i \times l_{ij}, \tag{1}$$

where l_{ij} is the horizontal distance between MU j and location i, i.e., $l_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$. Here, $\langle x_i, y_i \rangle$ and $\langle x_j, y_j \rangle$ are the $\langle \text{latitude}, \text{longitude} \rangle$ of location i and MU j, respectively. Thus, the 3-D distance between the DBS and MU j is $d_j = \sqrt{l_j^2 + h^2}$, where h is the DBS's altitude.

1) Average Pathloss Between a DBS and an MU: A probabilistic line-of-sight (LoS) channel is typically used to model the wireless access channel between an MU and the DBS. Let η_j^{LoS} and η_j^{NLoS} be the pathloss between MU j and the DBS in LoS and NLoS, respectively. We have [5]

$$\eta_j^{LoS} = 20 \log \left(\frac{4\pi f_0 d_j}{c} \right) + \xi^{LoS}, \tag{2}$$

$$\eta_j^{NLoS} = 20 \log \left(\frac{4\pi f_0 d_j}{c} \right) + \xi^{NLoS}, \tag{3}$$

where f_0 is the carrier frequency, ξ^{LoS} and ξ^{NLoS} are the average excessive pathloss in LoS and NLoS, respectively, c is the speed of light, and d_j is the distance between the DBS and MU j.

The probability of having LoS between the DBS and MU j (ρ_j) can be modeled as [5]

$$\rho_j = \frac{1}{1 + \alpha e^{-\beta(\theta_j - \alpha)}}, \tag{4}$$

where α and β are the two environmental parameters in the DSA, and θ_j is the elevation angle between the DBS and MU j (as indicated in Fig. 2), i.e.,

$$\theta_j = \arcsin \left(\frac{h}{d_j} \right). \tag{5}$$

The average pathloss between the DBS and MU j is [24]

$$\bar{\eta}_j = \rho_j \eta_j^{LoS} + (1 - \rho_j) \eta_j^{NLoS}. \tag{6}$$

2) Achievable Data Rate of a Wireless Access Link: Different MUs in the DSA will be assigned bandwidth to download data from the DBS. Denote b_j as the amount of bandwidth assigned to MU j. Thus, the achievable data rate of MU j is

$$r_j = b_j \log \left(1 + \frac{p 10^{-\bar{\eta}_j/10}}{N_0} \right), \tag{7}$$

where p is the transmission power density of the DBS and N_0 is the noise power density. Here, we assume the transmission power density p is fixed in different spectrum.

B. FSO Based Backhaul Link Communications Model

1) Received Optical Power: A working MBS transmits an optical beam carrying both data and energy to the DBS. The received optical power at the DBS (P^r) is [25][1]

$$P^r = P^t \eta^t \eta^w, \tag{8}$$

where P^t is the transmission power of the optical beam at the MBS, η^t is the coefficient to convert electrical power into optical power, and η^w is the environmental loss (caused by scattering and absorption of photons). Here, the environmental loss η^w in Eq. (8) can be estimated by [27]

$$\eta^w = 10^{-\frac{4.34 \sigma d^{fso}}{10}}, \tag{9}$$

where σ is the atmospheric attenuation coefficient, i.e., the amount of optical power loss in dB per kilometer, and d^{fso} is the distance between the DBS and the MBS, i.e.,

$$d^{fso} = \sqrt{\left(\sum\nolimits_{i \in \boldsymbol{I}} z_i \times \epsilon_i \right)^2 + (h - h^m)^2}. \tag{10}$$

Here, h^m is the altitude of the MBS, and ϵ_i is the horizontal distance between the MBS and location i. Thus, $\sum_{i \in \boldsymbol{I}} z_i \times \epsilon_i$ indicates the horizontal distance between the MBS and the DBS. Note that σ in Eq. (9) can be estimated by

$$\sigma = \frac{3.91}{v} \left(\frac{\lambda}{550} \right)^{-q}, \tag{11}$$

where λ is the wavelength of the optical beam, q is the size distribution of the scattering particles in the environment, and v is the visibility range of the environment (i.e., the maximum distance that an object can be clearly discerned). The visibility range (in km) depends on the weather conditions and can be obtained from the field tests. For example, when the weather is clear, v is normally larger than 50 km, but v is less than 1 km in a foggy weather. Note that the value of q is [27]

$$q = \begin{cases} 1.6, & v > 50, \\ 1.3, & 6 < v \leq 50, \\ 0.16v + 0.34, & 1 < v \leq 6, \\ v - 0.5, & 0.5 < v \leq 1, \\ 0, & v, \leq 0.5, \end{cases} \tag{12}$$

The optical beam received by the DBS is further separated into two portions. As illustrated in Fig. 3, one portion is used by

[1] Here, acquisition, pointing and tracking (APT) system [26] is presumed to be used in the architecture. Hence, we do not consider the pointing errors at an FSO receiver.

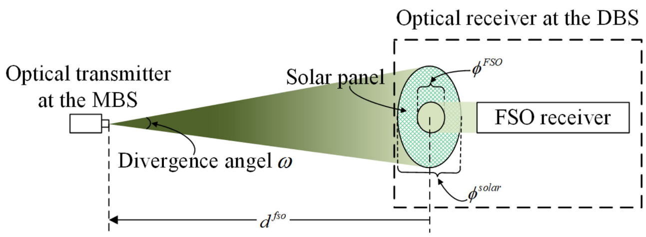

Fig. 3. Illustration of the received optical beam.

the solar panel, which converts the received optical power into electrical power to charge the battery of the DBS, and the other portion is used by the FSO receiver, which demodulates signals carried by the optical beam. Assume that the received optical power at the DBS is P^r (which can be calculated based on Eq. (8)). Let P^d and P^c be the amount of optical power used for retrieving data and charging the battery, respectively. Here,

$$P^r = P^d + P^c. \tag{13}$$

Note that, as shown in Fig. 3, the size of the hole in the solar panel could change P^c and P^d, i.e., a larger size of the hole indicates that a smaller portion of the received optical beam is received by the solar panel for charging the battery (i.e., smaller P^c) and a larger portion of the received optical beam is used to demodulate signals (i.e., larger P^d), and vice versa. Thus, we have

$$P^d = \left(\frac{\phi^{FSO}}{\phi^{solar}}\right)^2 P^r,\ P^c = \left(1 - \left(\frac{\phi^{FSO}}{\phi^{solar}}\right)^2\right) P^r, \tag{14}$$

where ϕ^{FSO} and ϕ^{solar} ($\phi^{FSO} < \phi^{solar}$) are the diameters of the lens for the FSO receiver (i.e., the diameter of the hole) and the solar panel, respectively.

2) FSO Based Backhaul Data Rate: The achievable data rate of the FSO based backhaul link is given by [28]

$$r^f = \frac{P^d}{E_p N_b}, \tag{15}$$

where N_b is the sensitivity of the FSO receiver in photons/bit, and E_p is the amount of energy carried by a single photon, i.e., $E_p = h_p c/\lambda$. Here, h_p is the Planck's constant, c is the light speed, and λ is the wavelength of the optical beam.

3) Optical Charging Model: Part of the received optical beam is used for charging batteries. Denote γ as the charging efficiency to convert the received optical power p^c into electrical power. Thus, the charging rate ($\hat{P}^c$) is

$$\hat{P}^c = \gamma P^c. \tag{16}$$

IV. Problem Formulation

In a drone assisted system, placing a DBS closer to the MBS may increase the DBS's hovering time and the backhaul link data rate but reduce the data rate between the DBS and the MUs, and vice versa. It is beneficial and nontrivial to find the optimal location of the DBS that can maximize the DBS's hovering time while guaranteeing the QoS of the MUs.

We assume that a number of MUs are located in a DSA, and different MUs have different data rate requirements. A DBS is placed over this DSA to provide services to these MUs. The energy consumption of the DBS comprises the communications and propulsion energy consumption. Normally, the propulsion system consumes much more energy than the communications system does in a DBS, and thus we only consider the propulsion energy consumption. The propulsion energy consumption of a DBS can be further separated into two parts, i.e., the energy consumption of the DBS traveling from the control station to the destination over the DSA and return to the control station (denoted as E^{fly}), and the DBS hovering energy consumption (denoted as E^{hover}). The value of E^{fly} depends on d^{fso}, i.e.,

$$E^{fly} = \xi d^{fso}, \tag{17}$$

where ξ indicates the average energy consumption of a drone by flying one km and d^{fso} is the flying distance, which can be calculated based on Eq. (10). The value of E^{hover} depends on the hovering power consumption P^{hover} and amount of hovering time T, i.e., $E^{hover} = TP^{hover}$. Denote $E^{battery}$ as the total amount of energy in the DBS's battery, and we assume that an optical link from a working MBS to a DBS (to transmit data and energy) can be established once the DBS is stably hovering at the destination. Then, we can derive the maximum DBS hovering time with optical charging as follows.

$$T = \frac{E^{battery} - \xi d^{fso}}{P^{hover} - \hat{P}^c}. \tag{18}$$

We define the DBS's extra hovering time (denoted as ΔT) as the hovering time of the DBS with optical charging minus that without the optical charging, i.e.,

$$\Delta T = \frac{E^{battery} - \xi d^{fso}}{P^{hover} - \hat{P}^c} - \frac{E^{battery} - \xi d^{fso}}{P^{hover}}. \tag{19}$$

We formulate the problem to determine the DBS location as well as the amount of bandwidth in the access links so as to maximize ΔT, i.e.,

$$\mathbf{P0}: \max_{z_i, h, b_j} \Delta T \tag{20}$$

$$\text{s.t.}: C1: \sum_{j \in \boldsymbol{J}} b_j \leq B, \tag{21}$$

$$C2: \sum_{i \in \boldsymbol{I}} z_i h_i^{\min} \leq h \leq h^{\max}, \tag{22}$$

$$C3: r_j \geq r_j^*, \tag{23}$$

$$C4: r^f \geq \sum_{j \in \boldsymbol{J}} r_j, \tag{24}$$

$$C5: \sum_{i \in \boldsymbol{I}} z_i = 1. \tag{25}$$

Here, C1 ensures that the bandwidth allocated to the MUs does not exceed the total amount of available bandwidth. C2 ensures the backhaul link is LoS (where $h_i^{\min}$ is the minimum altitude to ensure the LoS between the MBS and the DBS when the DBS is deployed over location i, and $h^{\max}$ is the maximum allowable altitude for the DBS). C3 indicates that each MU's data rate requirement should be guaranteed in the access link, where r_j^* is the data rate requirement of MU j. C4 ensures the achievable backhaul link data rate no less than the achievable access link data rates. C5 implies that the DBS can only be deployed over one location. Note that $\mathbf{P0}$ is a non-linear, mixed integer, and non-convex problem because 1) the objective of the problem, i.e., ΔT, is a non-convex function with respect to the location of the DBS (i.e., z_i and h), and 2) C4 in P0 is not a convex constraint with respect to z_i and h. Thus, it is nontrivial to derive the optimal solution of $\mathbf{P0}$.

V. Heuristic Algorithm

We propose a heuristic algorithm, i.e., TWIST, to efficiently solve the proposed problem $\mathbf{P0}$. The intuition of TWIST is to first find a feasible DBS placement that can satisfy all the constraints in $\mathbf{P0}$, and then look for a better DBS placement that reduces the distance to the MBS so as to increase ΔT. TWIST is summarized in Algorithm 1.

A. Initial 3-D DBS Placement

We sort all the locations in the DSA in the ascending order based on ϵ_i, i.e., the horizontal distance between location i and the MBS. Let $\boldsymbol{I'}$ be the set of the ordered locations, i.e., $\boldsymbol{I'} = \{i \in \boldsymbol{I} | \epsilon_i \leq \epsilon_{i+1}\}$. Also, the initial altitude of the DBS is assumed to be $\max\{h^m, h_i^{min}\}$, which incurs the shortest 3-D distance to the MBS among other possible altitudes. h^m (where $h_i^{max} \geq h^m, \forall i \in \boldsymbol{I}$) is the altitude of the working MBS outside the DSA, as illustrated in Fig. 2. The initial DBS placement is to iteratively deploy over location i in $\boldsymbol{I'}$ with the altitude equal to $\max\{h^m, h_i^{min}\}$ until a feasible 3-D DBS placement, which can satisfy Constraints C1, C2, C3, and C4 in $\mathbf{P0}$, has been found. Specifically,

1) Select location i in $\boldsymbol{I'}$ based on the ascending order of their horizontal distance. Deploy the DBS over location i with the altitude equal to $\max\{h^m, h_i^{min}\}$.
2) For each MU, calculate the pathloss from the DBS to the MU (i.e., $\bar{\eta}_j$) based on Eq. (6), and then calculate the amount of bandwidth required of an MU to satisfies its the minimum data rate requirement in C3, i.e., $b_j = r_j^* / \log(1 + \frac{p10^{-\bar{\eta}_j/10}}{N_0})$.
3) The achievable data rate of the FSO based backhaul link (i.e., r_i^f) is calculated based on Eq. (15).
4) If $\sum_{j\in\boldsymbol{J}} b_j \leq B$ (i.e., C1) and $r_i^f \geq \sum_{j\in\boldsymbol{J}} r_j^*$ (i.e., C4) can be satisfied, then we say that location i and its altitude $\max\{h^m, h_i^{min}\}$ are the initial optimal 3-D placement for the DBS, i.e., $i^* = i$ (i.e., $z_{i^*} = 1$) and $h^* = \max\{h^m, h_{i^*}^{min}\}$; otherwise, the next location in $\boldsymbol{I'}$ will be selected to check if it is a feasible DBS placement by executing from Steps 1–4.

Let $d_{i^*}^{fso}$ be the 3-D distance between the MBS and the DBS if the DBS is deployed in its initial 3-D placement (i.e., $\langle x_{i^*}, y_{i^*}, \max\{h^m, h_{i^*}^{min}\}\rangle$). Based on Eq. (10), we have

$$d_{i^*}^{fso} = \sqrt{(\epsilon_{i^*})^2 + (\max\{h^m, h_{i^*}^{min}\} - h^m)^2}, \tag{26}$$

Algorithm 1: TWIST Algorithm.

Initialize $\boldsymbol{I'}$.
for ($i = 1,\ i \leq |\boldsymbol{I'}|,\ i = i + 1$) {
  Set $x_i = 1$;
  $h = \max\{h^m, h_i^{min}\}$;
  $\forall j \in \boldsymbol{J}$, calculate $\bar{\eta}_j$, b_j, and r_i^f;
  if $\sum_{j\in\boldsymbol{J}} b_j \leq B$ *and* $r_i^f \geq \sum_{j\in\boldsymbol{J}} r_j^*$ **then**
    $i^* = i$, $z_{i^*} = 1$, and $h^* = \max\{h^m, h_{i^*}^{min}\}$;
  else
    $z_i = 0$;
  end
}
Calculate $d_{i^*}^{fso}$ based on Eq. (26).
$k = i^*$.
for ($i = k - 1,\ i \geq 1,\ i = i - 1$) {
  $x_i = 1$;
  for ($h_i = h_i^{min}, h_i \leq \min\left\{\sqrt{\left(d_{i^*}^{fso}\right)^2 - \epsilon_i{}^2}, h^{\max}\right\},\ h = h + \Delta h$) {
    $\forall j \in \boldsymbol{J}$, calculate $\bar{\eta}_j$ and b_j;
    Calculate r_i^f based on Eq. (15);
    if $\sum_{j\in\boldsymbol{J}} b_j \leq B$ *and* $r_i^f \geq \sum_{j\in\boldsymbol{J}} r_j^*$ **then**
      $\boldsymbol{H}_i = \boldsymbol{H}_i \cap h_i$;
    end
  }
  if $\boldsymbol{H}_i = \emptyset$ **then**
    $z_i = 0$;
  else
    $\bar{h}_i = \arg\min_{h_i \in \boldsymbol{H}_i} |h_i - h^m|$;
    Calculate $d_i^{fso} = \sqrt{(\epsilon_i)^2 + (\bar{h}_i - h^m)^2}$;
    if $d_i^{fso} < d_{i^*}^{fso}$ **then**
      $z_{i^*} = 0$, $i^* = i$, $z_{i^*} = 1$, and $h^* = \bar{h}_i$;
    else
      $z_i = 0$;
    end
  end
}

where ϵ_{i^*} is the horizontal distance between the MBS and location i^* .

B. Recursive Search

Based on Eq. (19), it is easy to derive that reducing d^{fso} could increase ΔT. Thus, recursive search is to backtrack the locations whose indices are lower than i^* in $\boldsymbol{I'}$ in order to find a better 3-D placement that could reduce d^{fso} while satisfying all the constraints in $\mathbf{P0}$. That is, the DBS is iteratively deployed over one of these locations and adjust the altitude of the DBS to find a better 3-D placement. Specifically,

1) Let $k = i^*$ and we iterively select a location in $\boldsymbol{I'}$ according to the order of $\langle k-1, k-2, \ldots, 1\rangle$. Denote i as the selected location in the current iteration.

2) Deploying the DBS over location i could be a better 3-D placement if d_i^{fso} is reduced, i.e., $\sqrt{{\epsilon_i}^2 + (h_i - h^m)^2} < d_{i^*}^{fso}$, where $d_{i^*}^{fso}$ is derived from Eq. (26). Thus, we have $h_i \leq \sqrt{(d_{i^*}^{fso})^2 - \epsilon_i^2} + h^m$. The next step (i.e., Step 3)) is to find all the possible values of h_i to satisfy all the constraints in **P0**, where $h_i^{\min} \leq h_i \leq \min\{\sqrt{(d_{i^*}^{fso})^2 - {\epsilon_i}^2} + h^m, h^{\max}\}$. Note that if $h_i^{\min} > \sqrt{(d_{i^*}^{fso})^2 - \epsilon_i^2} + h^m$, then deploying the DBS over location i cannot be a better 3-D placement.
3) Denote Δh as the step size of adjusting the altitude. we will iteratively select the value of h_i from the order $\{h_i^{\min}, h_i^{\min} + \Delta h, h_i^{\min} + 2\Delta h, \ldots\}$ until $h_i > \min\{\sqrt{(d_{i^*}^{fso})^2 - {\epsilon_i}^2},\ h^{\max}\}$. In each iteration, check whether the selected h_i is a feasible solution to meet all the constraints or not. If h_i is a feasible solution, put it into set $\boldsymbol{H}_i$. After finishing all the iterations, if $\boldsymbol{H}_i = \emptyset$, then deploying the DBS over location i is not a better DBS placement; otherwise, find the altitude $\bar{h}_i$ that incurs the shortest distance to the MBS among other altitudes in $\boldsymbol{H}_i$, i.e., $\bar{h}_i = \arg\min_{h_i \in \boldsymbol{H}_i} |h_i - h^m|$. Calculate d_i^{fso} based on Eq. (26), where d_i^{fso} is the 3-D distance between the MBS and the DBS when the DBS is placed over location i with the altitude equal to $\bar{h}_i$. If $d_i^{fso} < d_{i^*}^{fso}$, the optimal DBS placement becomes $i^* = i$ (i.e., $z_{i^*} = 1$) and $h^* = \bar{h}_i$.
4) Iterively select the next location in $\boldsymbol{I}'$ by executing Steps 1-3 to update the the optimal DBS placement $\langle i^*, h^* \rangle$ until all the locations have been selected.

VI. Simulation Results

The performance of TWIST is evaluated via simulations by comparing it with two other baseline algorithms, i.e., CLP (center location placement) [29] and MTP (maximal throughput placement) [30]. In CLP, the DBS is first deployed at the center of all the MUs, and then the altitude of the DBS and the bandwidth assignment are adjusted to satisfy the MUs' data rate requirements in the access links. The intuition of MTP is to derive the optimal DBS placement such that the aggregated achievable data rate of the MUs in downloading data from the MBS via the DBS is maximized. Both algorithms assume that FSO is applied in backhaul communications.

In the simulation, the FSO transmission power is 1200W [31].[2] Assume that a DSA forms a rectangle area with range of $\langle 0 \sim 1\,\text{km}, 0 \sim 1\,\text{km}\rangle$. The DSA is further partitioned into 40 000 locations, and the size of each location is 5 m $\times$ 5 m. The working MBS is placed at $\langle -100\,\text{m}, -100\,\text{m}\rangle$. Initially, there are 40 MUs uniformly distributed in the DSA. The data rate requirements of these MUs follow a normal distribution with the a mean and standard deviation equal to 2 Mbps and 1 Mbps, respectively. The diameters of the solar panel and the FSO receiver are 0.1 m and 0.06 m, respectively. Other simulation parameters are listed in Table I.

TABLE I
Simulation Parameters

FSO transmission power [31]	1200 W
Noise power spectral density (N_0)	-104 dBm/Hz
Solar panel diameter (ϕ^{solar})	0.1 m
FSO receiver diameter (ϕ^{FSO})	0.06 m
Receiver sensitivity (N_b) [32]	$2 \times 10^5 photons/bit$
FSO wavelength (λ)	1550 nm
Environment index (β) [33]	0.16
Environment index (α) [33]	9.61
FSO charging efficiency (γ)	20%
Maximum altitude of DBS (h^{max})	300 m
The altitude of working MBS (h^m)	20 m
DBS power consumption (P^{hover})	1000 W
DBS battery capacity ($E^{battery}$)	30000 mAh $\times$ 20 V
Carrier frequency (f_0)	2 GHz
DBS transmission power (p)	0.5 W/kHz

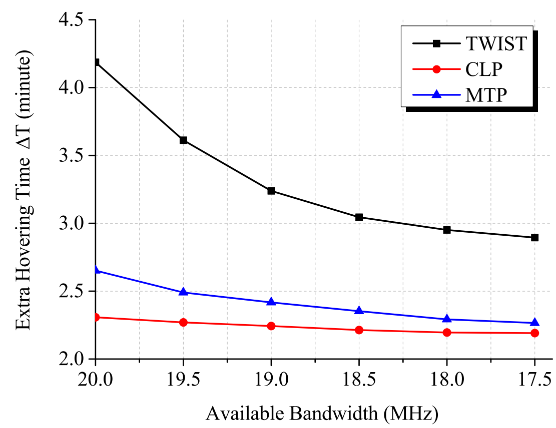

Fig. 4. Extra hovering time by varying the amount of available bandwidth.

Fig. 4 shows the extra hovering time for the three algorithms when the amount of available bandwidth in the access links is varying from 20 MHz to 17.5 MHz. The visibility range is $v = 0.9$ km. From the figure, we can find that TWIST always achieves the longest extra hovering time as compared to MTP and CLP. Also, the extra hovering time incurred by the three algorithms increases as the amount of available bandwidth in the access links decreases. This is because as the amount of available bandwidth decreases, each MU can be allocated with less bandwidth in the access link, and thus the DBS needs to be placed over a location that incurs better channel gain to the MUs in order to satisfy their data rate requirements. The location incurring better channel gain to the MUs in the access links normally leads to a longer distance in the backhaul link, and thus reduces ΔT. For example, Fig. 5 exhibits the placements of the DBS acquired by the three algorithms. Here, the locations with orange and blue color imply the DBS locations (incurred by the three algorithms) when the amount of available bandwidth are 18 MHz and 20 MHz, respectively. From the figure, we can find that as the available bandwidth changes from 20 MHz to 18 MHz, the locations of the DBSs incurred by the three

[2]In order to alleviate the safety issue of using high power FSO, an ATP (Acquisition, Tracking, and Pointing) system could be applied such that the optical beam can be accurately point to the solar panel of the DBS. Another possible solution to alleviate the safety issue of using high power FSO is to use several low power optical beams, rather than a single high optical beam, to convey data and energy to the DBS.

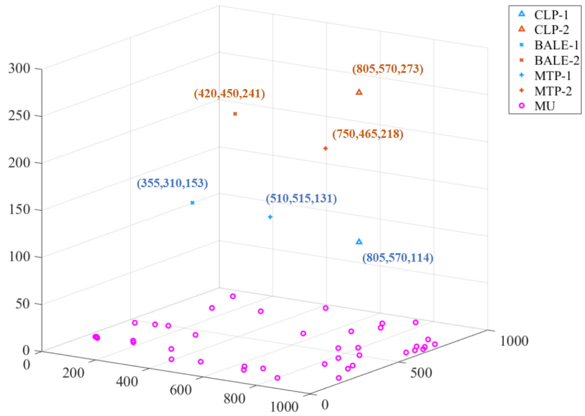

Fig. 5. Location of the DBS with different available bandwidth.

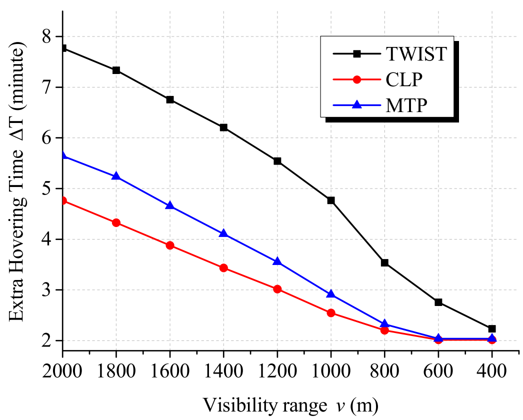

Fig. 6. Extra hovering time by varying visibility range.

algorithms moves farther away from the MBS, which reduces the charging rate $\hat{P}^c$, thus reducing the extra hovering time. Note that, as shown in Fig. 4, the difference of the extra hovering time between TWIST and MTP/CLP increases as the amount of available bandwidth in the access links decreases.

Fig. 6 shows how the visibility range v affects ΔT. As mentioned before, the visibility range v is normally determined by the weather conditions in the area. The available bandwidth in the access links is 20 MHz. A larger v results in a higher charging rate and backhaul link capacity. From the figure, we can see that the extra hovering time for the three algorithms decreases when v reduces because the charging rate decreases as v reduces. Meanwhile, the difference of the extra hovering time between TWIST and MTP/CLP also reduces when v reduces. This is because, although the DBS placement incurred by TWIST has shorter backhaul link than that incurred by CLP MTP, the amount of charging rate gain (which is caused by a shorter backhaul link) reduces as v reduces.

Assume that the available bandwidth of the access network is 20 MHz and the number of MUs is 40. Fig. 7 shows the extra hovering time of the DBS by varying the size of the DSA. When the size of the DSA is 0.5 km $\times$ 0.5 km, the extra hovering time incurred by TWIST is much higher than MTP and CLP. As the size of the DSA increases, the extra hovering time incurred by three algorithms becomes similar because, as the size of DSA

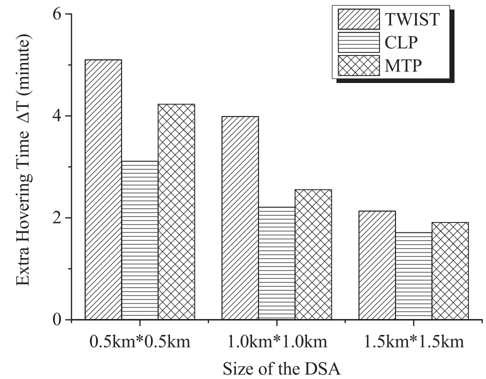

Fig. 7. Extra hovering time in different disaster area size.

increases, the distribution of the MUs (which are uniformly distributed in the DSA) are more sparse. That is, the distance between two MUs becomes larger as the size of DSA increases. As a result, the average pathloss between the DBS and these MUs increases. In order to satisfy the data rate requirements in the access network, TWIST has to deploy the DBS closer to the MUs (to reduce the pathloss). Note that TWIST performs like CLP/MTP as the size of DSA increases, i.e., all the three algorithms try to find a feasible DBS placement to satisfy the data rate requirements.

VII. Conclusion

This work leverages the SoarNet architecture to achieve simultaneous data and energy transmission to the DBS in emergency communications. TWIST has been designed to determine the DBS placement and the bandwidth assignment such that the extra hovering time can be maximized. Extensive simulations have demonstrated that TWIST achieves the largest extra hovering time as compared to the other two baseline methods.

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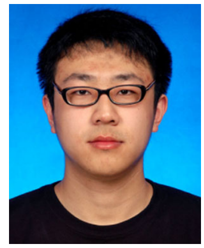
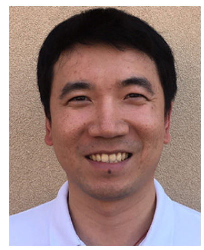


Di Wu (Student Member, IEEE) received the B.E. degree in communication engineering and the M.E. degree in information and communication engineering from the Harbin Institute of Technology, China, and the Ph.D. degree in electrical engineering from the New Jersey Institute of Technology (NJIT), in 2020. His research interests include mobile edge computing, network security, and drone assisted network.



Xiang Sun (Member, IEEE) received the B.E. and M.E. degrees from the Hebei University of Engineering, in 2008 and 2011, respectively, and the Ph.D. degree in electrical engineering from the New Jersey Institute of Technology (NJIT), in 2018. He is an Assistant Professor with Electrical and Computer Engineering with the University of New Mexico. His research interests include mobile edge computing, Internet of Things, wireless networks, and big-data-driven networking. He was the recipient of the several honors and awards, including the 2016 IEEE International Conference on Communications Best Paper Award, the 2017 IEEE Communications Letters Exemplary Reviewers Award, the 2018 NJIT Hashimoto Price, the 2018 Inter Digital Innovation Award on IoT Semantic Mashup, the 2019 NJIT Outstanding Doctoral Dissertation Award, and the 2019 IEICE Communications Society Best Tutorial Paper Award. He has served as a TPC member for many conferences. He is currently an Associate Editor of *Digital Communications and Networks*.

Nirwan Ansari (Fellow, IEEE) received the Ph.D. degree from Purdue University, the MSEE degree from the University of Michigan, and the BSEE (*summa cum laude* with a perfect GPA) degree from the New Jersey Institute of Technology (NJIT).

He authored Green Mobile Networks: A Networking Perspective (Wiley-IEEE, 2017) with T. Han, and coauthored two other books. He has also coauthored more than 600 technical publications, more than310 published in widely cited journals/magazines. He was the Guest-Edited of the number of special issues covering various emerging topics in communications and networking. He has served on the editorial/advisory board of over ten journals including as an Associate Editor-in-Chief of *IEEE Wireless Communication Magazine* His current research focuses on green communications and networking, cloud computing, drone-assisted networking, and various aspects of broadband networks. He is also a Fellow with the National Academy of Inventors.

He was elected to serve in the IEEE Communications Society (ComSoc) Board of Governors as a Member-at-large, is current Director of ComSoc Educational Services Board, has chaired some ComSoc technical and steering committees, has been serving in many committees such as the IEEE Fellow Committee, and has been actively organizing numerous IEEE International Conferences/Symposia/Workshops. He is frequently invited to deliver keynote addresses, distinguished lectures, tutorials, and invited talks. Some of his recognitions include several Excellence in Teaching Awards, a few Best Paper Awards, the NCE Excellence in Research Award, several ComSoc TC technical recognition awards, the NJ Inventors Hall of Fame Inventor of the Year Award, the Thomas Alva Edison Patent Award, Purdue University Outstanding Electrical and Computer Engineering Award, the NCE-100 Medal, and designation as a COMSOC Distinguished Lecturer. He has also been granted more than 40 U.S. patents.